

Data Cleaning

What is data cleaning

Data cleaning is the process of fixing or removing incorrect, corrupted, incorrectly formatted, duplicate, or incomplete data within a dataset. When combining multiple data sources, there are many opportunities for data to be duplicated or mislabeled. If data is incorrect, outcomes and algorithms are unreliable, even though they may look correct.

Data cleaning is the process of removing incorrect, duplicate, or otherwise erroneous data from a dataset. These errors can include incorrectly formatted data, redundant entries, mislabeled data, and other issues; they often arise when two or more datasets are combined together. Data cleaning improves the quality of your data as well as any business decisions that you draw based on the data.

Data cleaning is the process of detecting incomplete, incorrect, inaccurate, or irrelevant parts of the data and then replacing, modifying, or deleting the dirty data

Data cleaning is also called data cleansing

Steps of Data cleaning

- Import dataset
- Merge datasets into one datasets
- Identify error
- Standardise process
- Scrub for duplicates
- Validate accuracy

Data Cleaning

Handling missing values

What is missing values & method to handle missing values

In the variable or any observation, values are not stored called as missing values/data

Age	Sex	Male	Female	Age range
23	Male		0	21-30
12	Female	0	1	
34	Male	1	0	31-40
26		0	1	21-30
	Male	1	0	11-20
38	Male	1	0	31-40

Method to handle missing values

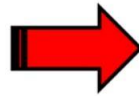
1. Ignore missing values row/delete rows
2. Fill missing value manually
3. Global constant
4. Measure of central tendency (mean , median, mode)
5. Measure of central tendency for each class
6. Most probable values

Ignore missing values row/delete rows

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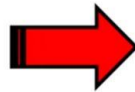
Dev	Exper.	Salary
Java	1	20000
Python	4	
Java	1.5	25000
Python	2	40000
Python	4	80000
	2	



Dev	Exper.	Salary
Java	1	20000
Java	1.5	25000
Python	2	40000
Python	4	80000

Fill missing values manually

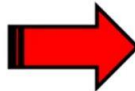
St. Percentage	Present
80	P
45	P
34	
00	A



St. Percentage	Present
80	P
45	P
34	P
00	A

Global constant

Age	Sex
23	Male
12	Female
34	Male
26	
	Male
38	Male



Age	Sex
23	Male
12	Female
34	Male
26	Unknown
0	Male
38	Male

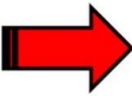
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Measure of central tendency

Mean , median,mode

Dev	Exper.	Salary
Java	1	20000
Python	4	
Java	1.5	25000
Python	2	40000
Python	4	80000
Java	2	

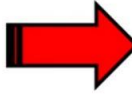


Dev	Exper.	Salary
Java	1	20000
Python	4	41250
Java	1.5	25000
Python	2	40000
Python	4	80000
Java	2	41250

Most probable value

Use ml algorithms – Linear Regg,decision tree

Dev	Exper.	Salary
Java	1	20000
Python	4	
Java	1.5	25000
Python	2	40000
Python	4	80000
Java	2	



Dev	Exper.	Salary
Java	1	20000
Python	4	80000
Java	1.5	25000
Python	2	40000
Python	4	80000
Java	2	30000

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Practical - Ignore missing values row/delete rows

First step – gether dataset & import library

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

load dataset & find shape of data set

```
df=pd.read_csv('train.csv')
df.shape
```

check the dataset using head

```
df.head(6)
```

there is more column in this dataset but visible only some.

Use **set_option** function to visible all

```
pd.set_option('display.max_columns',None)
pd.set_option('display.max_rows',None)
```

Using **head function** to show all

```
df.head(6)
```

for information of dataset

```
df.info()
```

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for check missing data

```
df.isnull()
```

this function display True and False.

If data is missing it indicates true otherwise false.

but we want to sum of total missing values in particular column

```
df.isnull().sum()
```

for visualize this data

use heatmap to visualise data

```
plt.figure(figsize=(25,25))  
sns.heatmap(df.isnull())
```

in this plot you can see where white color indicate null values and black color indicate the value is present

find percentage of missing data in each column

use total missing data in column / total no of data *100

```
null_var=df.isnull().sum()/df.shape[0]*100  
null_var
```

we can remove data from this dataset if more than 20% of data is missing we should remove that column find this.

We have store percentage of missing data in null_var

```
drop_columns=null_var>null_var>17].keys()  
drop_columns
```

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now we need to drop this column where more then 17% data is missing
use drop() of dataframe and store in another frame

```
df2_drop_columns=df.drop(columns=drop_columns)  
df2_drop_columns
```

now find shpe of the new dataset

```
df2_drop_columns.shape
```

now you can find remove this dataset column

now can visualize new dataset through heatmap

```
snp.heatmap(df2_drop_columns.isnull())
```

now we have to remove those missing values in rows also

use dropna()

```
df3_drop_rows=df2_drop_columns.dropna()
```

now find shape of df

```
df3_drop_rows.shape
```

now again visualize dataset using heatmap is there any null values of
not

```
plt.figure(figsize=(25,25))  
snp.heatmap(df3_drop_rows.isnull())
```

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now you can check there is no any missing values in this dataset

if you want in number format for any missing values

```
df3_drop_rows.isnull().sum()
```

now we need to check the data is cleaned or not we need to see data distribution using histogram

seaborn distplot only using on values . so now we need to find out which columns has a value using **select_dtypes()**

to find out columns name of value

```
df3_drop_rows.select_dtypes(include=['int64', 'float64']).columns
```

to select any one column and check distplot or histogram

```
sns.distplot(df['MSSubClass'])
```

this graph is original datachart graph

we need to check on cleaned datachart graph

```
sns.distplot(df3_drop_rows['MSSubClass'])
```

we can see new graph is relevancy to original datagraph

now if we want to show exact distribution,

need to overlap graph for both dataset

```
sns.distplot(df['MSSubClass'])  
sns.distplot(df3_drop_rows['MSSubClass'])
```


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we can clearly see there is no or slightly difference on data distribution compare to original data set

that means we have right way to cleaned the data..

this is only single column distribution we need to check all the columns data distribution

we have all column in new dataframe where only value available

there is total 36 columns has value

for all distribution data compare we need enumerate object class for create subplot.

```
plt.figure(figsize=(25,25))
for i,v in enumerate(num_var):

    plt.subplot(9,4,i+1)
    sns.distplot(df[v],bins=20)
    sns.distplot(df3_drop_rows[v],bins=20)
```

bins is optional you can remove also

there is 36 distplot using data distribution

now we need to check categorical data data distribution

first we need to find categorical data columns

```
df3_drop_rows.select_dtypes(include=['object']).columns
```

there is list of colums available for categorical data.

Now we have to check distribution for both original dataset and cleaned dataset

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Now check how much off data category available in each dataset

First randomly select any columns and counts the total value

```
df['MSZoning'].value_counts()
```

there is some data with category and find percentage of original dataset for particular category

```
df['MSZoning'].value_counts()/df.shape[0]*100
```

now find cleaned data percentage of category

```
df['MSZoning'].value_counts()/df.shape[0]*100,  
df3_drop_rows['MSZoning'].value_counts()/df3_drop_rows.shape[0]*100
```

for perfection use concat function to clear view

```
pd.concat([df['MSZoning'].value_counts()/df.shape[0]*100,  
          df3_drop_rows['MSZoning'].value_counts()/df3_drop_rows.shape[0]*100],axis=1,keys=['MSZoning_original','MSZoning_new'])
```